

Numbered Increasing $(1, 2)$ -Trees and the Euler Zigzag Numbers

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Some background

The *Euler numbers*, or *Euler zigzag numbers*, are the sequence

$$A_n = 1, 1, 1, 2, 5, 16, 61, 272, \dots$$

They satisfy the recurrence

$$2A_{n+1} = \sum_{k=0}^n \binom{n}{k} A_k A_{n-k}.$$

This comes up a lot in combinatorics.

The objects we will count are the following.

Definition 1 (Numbered increasing $(1, 2)$ -trees). A *numbered increasing $(1, 2)$ -tree* is an increasing rooted tree such that every non-endpoint vertex has one or two children. Two trees on the same number of vertices are considered the same (isomorphic) if one can be reoriented so as to recover the other. The trees are numbered (labeled $1, 2, \dots, n$) such that the labels strictly increase along every path from the root to a leaf.

The notes illustrate this with two small examples: a pair of trees that are considered the same, since one can be rearranged to match the other, and a pair of trees that are distinct, since no such rearrangement exists.

We write

$$f(n) = \text{the number of distinct numbered increasing } (1, 2)\text{-trees on } n \text{ vertices.}$$

Lemma 1.

$$\sum_{n \geq 0} E_n \frac{x^n}{n!} = \sec x + \tan x.$$

Main result

Proposition 1. For each n , $f(n) = E_n$.

Before the proof, we check the first few values directly:

$$f(1) = 1, \quad f(2) = 1, \quad f(3) = 2, \quad f(4) = 5,$$

which agree with E_1, E_2, E_3, E_4 .

Proof. Let

$$y = \sum_{n \geq 1} f(n) \frac{x^n}{n!} = x + \frac{x^2}{2!} + 2 \frac{x^3}{3!} + \cdots.$$

Taking the derivative with respect to x and reindexing,

$$y' = \sum_{n \geq 1} f(n) \frac{n x^{n-1}}{n!} = \sum_{n \geq 0} f(n+1) \frac{x^n}{n!}.$$

So y' is the exponential generating function counting numbered increasing $(1, 2)$ -trees on $n+1$ vertices. We split such a tree into cases according to the structure at the root:

- (a) a single vertex ($n = 0$);
- (b) the tree has one subtree of the root, which is an increasing $(1, 2)$ -tree with n vertices;
- (c) the tree has two subtrees of the root, each of which is an increasing $(1, 2)$ -tree, with n vertices in all.

From these cases we obtain

$$y' = 1 + y + \frac{1}{2}y^2.$$

(This checks out if you compute it, but we will not carry out that computation here.)

This differential equation has the solution

$$y = \sec x + \tan x - 1.$$

To see why, compute the derivative

$$y' = \sec x \tan x + \sec^2 x,$$

and substitute y into the right-hand side of the equation:

$$\begin{aligned} 1 + y + \frac{1}{2}y^2 &= 1 + (\sec x + \tan x - 1) + \frac{1}{2}(\sec x + \tan x - 1)^2 \\ &= (\sec x + \tan x) + \frac{1}{2}(\sec^2 x + \tan^2 x + 1 + 2 \sec x \tan x - 2 \sec x - 2 \tan x) \\ &= \sec x \tan x + \frac{1}{2}(\sec^2 x + \tan^2 x + 1) \\ &= \sec x \tan x + \frac{1}{2}(\sec^2 x + \sec^2 x) \\ &= \sec x \tan x + \sec^2 x \\ &= y', \end{aligned}$$

where we used $\tan^2 x + 1 = \sec^2 x$. This is exactly what we wanted.

Therefore

$$y = \sum_{n \geq 1} E_n \frac{x^n}{n!},$$

which is the generating function of Lemma 1 with its constant term removed. Matching coefficients gives

$$f(n) = E_n \quad (\text{starting from } E_1).$$

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